

Complete Real-Root Classification for the Domination Polynomials of Square Book Graphs

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Abstract

Let B_n be the graph formed from n four-cycles sharing a common edge, and let $D_n(x)$ be its domination polynomial. For every integer $n \geq 2$ we determine all real zeros, with algebraic multiplicity. Zero has multiplicity two. If n is even, there is one simple zero in $(-\infty, -2)$ and one in $(-1/2, 0)$. If n is odd, there are two simple zeros in $(-1/2, 0)$. There are no others. The proof combines two rational changes of variable with sharp coefficient-variation bounds: a monotone normalized coefficient sequence in the even case and a log-concave quotient in the odd case. The restriction $n \geq 2$ is essential to the formulation: B_1 has no nonzero real domination zero.

Keywords. domination polynomial; book graph; real roots; Descartes' rule of signs; log-concavity.

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1 Statement and context

For a finite graph G , write

$$D(G, x) = \sum_{\substack{S \subseteq V(G) \\ N[S] = V(G)}} x^{|S|}$$

for its ordinary domination polynomial. For $n \geq 1$, let B_n have vertices

$$u, v, a_1, b_1, \dots, a_n, b_n$$

and exactly the edges $uv, ua_i, a_i b_i, b_i v$ for $1 \leq i \leq n$. Thus B_n consists of n copies of C_4 sharing the edge uv . Put $D_n(x) = D(B_n, x)$.

Alikhani, Brown, and Jahari obtained a closed formula for D_n and posed a broader question about the domination roots of square book graphs [1, Theorem 2.9 and Question 3.3]. Jahari and Alikhani also studied the same formula and the limiting locus of the complex roots [2, Theorems 5 and 10]. More recently, Rather proved the existence of one real root in each of two specified intervals when n is even and formulated the exact parity-dependent count as a conjecture [3, Proposition 5.3 and Conjecture 5.4]. The theorem below supplies the corresponding all- n upper bounds, the odd-case existence, and the multiplicity assertions for the explicitly stated range $n \geq 2$. This source comparison is bounded to the works just cited; no claim of exhaustive literature coverage or global priority is made.

For adjacent methodological context only, exact coefficient arguments and finite certification also appear in an unrelated enumerative probability problem of Zeng [4]. That paper concerns cycle lengths of permutations, not domination polynomials, and supplies no premise for the result proved here.

Theorem 1. *For every integer $n \geq 2$, zero is a root of D_n of algebraic multiplicity exactly two. In addition:*

1. *if n is even, D_n has exactly one simple real root in $(-\infty, -2)$ and exactly one simple real root in $(-1/2, 0)$;*
2. *if n is odd, D_n has exactly two simple real roots in $(-1/2, 0)$.*

There are no other nonzero real roots. Hence the total real-root multiplicity is four for every $n \geq 2$.

The argument uses Descartes' rule in the form that the number of positive real roots of a nonzero real polynomial, counted with algebraic multiplicity, is at most the number of sign variations in its coefficient sequence after zero coefficients are deleted.

2 The polynomial and the middle interval

Partition the dominating sets according to their intersection with $\{u, v\}$. If both central vertices are selected, the contribution is $x^2(1+x)^{2n}$. If exactly one is selected, every pair $\{a_i, b_i\}$ must contribute a nonempty subset, giving $2x(x^2+2x)^n$. If neither is selected, every pair must again contribute a nonempty subset; the all- a and all- b selections are precisely the two choices that fail to dominate one of the central vertices. Their combined contribution is $(x^2+2x)^n - 2x^n$. Therefore

$$D_n(x) = (x^2 + 2x)^n(2x + 1) + x^2(1 + x)^{2n} - 2x^n. \quad (1)$$

For $n \geq 2$, define the polynomial

$$Q_n(x) = \frac{D_n(x)}{x^2} = (1+x)^{2n} + x^{n-2}((x+2)^n(2x+1) - 2). \quad (2)$$

Here $Q_2(0) = 3$, while $Q_n(0) = 1$ for $n \geq 3$. Thus zero has multiplicity exactly two. Moreover, $D_n(x) > 0$ for $x > 0$, since its coefficients count dominating sets and are nonnegative, so there is no positive real root.

We next exclude the closed interval $[-2, -1/2]$. Put $y = -x$. For $1/2 \leq y \leq 2$, equation (2) becomes

$$Q_n(-y) = (1-y)^{2n} + (-1)^n y^{n-2}((2-y)^n(1-2y) - 2). \quad (3)$$

The bracket is at most -2 . Hence (3) is positive when n is odd. If n is even, then $|1-y| \leq 1$ and

$$0 \leq \frac{(1-y)^2}{y} \leq \frac{1}{2},$$

where the upper bound is equivalent to $(y-1/2)(y-2) \leq 0$. For $n > 2$ this gives

$$\frac{(1-y)^{2n}}{y^{n-2}} = (1-y)^4 \left(\frac{(1-y)^2}{y} \right)^{n-2} \leq 1.$$

For $n = 2$ the quotient is simply $(1-y)^4 \leq 1$. Dividing (3) by the positive number y^{n-2} now yields $Q_n(-y) \leq -y^{n-2} < 0$. Consequently no root occurs in the closed middle interval, including either endpoint.

3 The interval to the left of -2

For $t > 0$, substitute $x = -2 - t$ in (2) and write

$$\begin{aligned} T_n(t) &:= Q_n(-2 - t) = H_n(t) - 2(-1)^n(t+2)^{n-2}, \\ H_n(t) &:= (t+1)^{2n} - t^n(t+2)^{n-2}(2t+3). \end{aligned} \quad (4)$$

Since $(t+1)^2 = 1 + t(t+2)$, expansion of the first term and combination of the last summand give

$$H_n(t) = \sum_{r=0}^{n-1} \binom{n}{r} t^r (t+2)^r + t^n (t+2)^{n-2} (t+1)^2. \quad (5)$$

Every coefficient in (5) is nonnegative, and every degree from 0 through $2n$ has a positive coefficient. Thus, for odd n , all coefficients of T_n are positive and there is no positive root. This excludes $(-\infty, -2)$ in the odd case.

Suppose now that n is even. The subtracted term in (4) affects only degrees $0 \leq k \leq n-2$. Since the second term in the definition of H_n begins at degree n , the affected coefficients are

$$[t^k]T_n = \binom{2n}{k} - 2^{n-1-k} \binom{n-2}{k}, \quad 0 \leq k \leq n-2. \quad (6)$$

The ratio of the first positive term in (6) to the second has consecutive quotient

$$\frac{\rho_{k+1}}{\rho_k} = \frac{2(2n-k)}{n-2-k} > 1 \quad (0 \leq k < n-2).$$

Hence the affected coefficients, after zeros are deleted, consist of a negative block followed by at most one positive block. The constant coefficient is $1 - 2^{n-1} < 0$, while all coefficients of degree at least $n-1$ are positive by (5). Thus T_n has exactly one coefficient sign variation. (For $n=2$ the affected range is only $k=0$, so the ratio comparison is vacuous.) Descartes' rule gives at most one positive root, counted with multiplicity. Because $T_n(0) < 0$ and its leading coefficient is 1, continuity supplies such a root. It is therefore unique and simple. This proves the even-case assertion on $(-\infty, -2)$.

4 A coefficient certificate near zero

The substitution

$$x = -\frac{t}{2(1+t)}$$

maps $t > 0$ bijectively onto $(-1/2, 0)$ and has nonzero derivative there. After clearing the positive denominator in (2), put

$$\begin{aligned} P_n(t) &:= [2(1+t)]^{2n} Q_n\left(-\frac{t}{2(1+t)}\right) = A_n(t) + (-1)^n(C_n(t) - B_n(t)), \\ A_n(t) &= (t+2)^{2n}, \\ B_n(t) &= 2^{n+3}t^{n-2}(1+t)^{n+2}, \\ C_n(t) &= 4t^{n-2}(1+t)(3t+4)^n. \end{aligned} \quad (7)$$

All coefficients below degree $n - 2$ come only from A_n and are positive. For degree $k = n - 2 + j$, where $0 \leq j \leq n + 2$, let $b_j = [t^k]B_n = 2^{n+3} \binom{n+2}{j} > 0$. Normalizing the other two contributions by b_j gives

$$a_j := \frac{[t^k]A_n}{b_j} = \frac{(2n)!j!}{2^{j+1}(n+2)!(n-2+j)!}, \quad (8)$$

$$r_j := \frac{[t^k]C_n}{b_j} = 2^{n-1} \left(\frac{3}{4}\right)^j \frac{(n+2-j)(n+1+j/3)}{(n+2)(n+1)}. \quad (9)$$

These formulas include $r_{n+2} = 0$. They follow directly by expanding (7); for instance, the two possible contributions from C_n are

$$4 \left\{ \binom{n}{j} 3^j 4^{n-j} + \binom{n}{j-1} 3^{j-1} 4^{n-j+1} \right\},$$

with an out-of-range binomial coefficient understood as zero.

4.1 Even n

The normalized coefficient at degree $n - 2 + j$ is $a_j + r_j - 1$. From (8),

$$\frac{a_{j+1}}{a_j} = \frac{j+1}{2(n-1+j)} \leq \frac{1}{2},$$

so a_j strictly decreases. If $F_j = (n+2-j)(n+1+j/3)$, then

$$F_j - F_{j+1} = \frac{2(n+j+1)}{3} > 0,$$

and the geometric factor in (9) also decreases. Hence r_j strictly decreases through its terminal value zero, and $a_j + r_j - 1$ is strictly decreasing. Together with the positive lower-degree coefficients, this permits at most one sign variation.

The constant coefficient of P_n is 48 when $n = 2$ and 2^{2n} when $n > 2$, whereas its leading coefficient is $1 - 2^{n+3} < 0$. Therefore the coefficient sequence has exactly one sign variation. Descartes' rule and continuity give exactly one positive root, and the multiplicity bound makes it simple. This is the required root in $(-1/2, 0)$.

4.2 Odd n

Now $n \geq 3$, and the normalized coefficient is $1 + a_j - r_j$. On $0 \leq j \leq n + 1$, the sequence r_j is positive and log-concave: apart from a positive constant, (9) is the product of a geometric sequence and the two positive affine sequences $n + 2 - j$ and $n + 1 + j/3$. Each affine sequence is log-concave, and products of positive log-concave sequences are log-concave.

On the other hand, the ratios a_{j+1}/a_j increase, since the difference between consecutive ratios is

$$\frac{n-2}{2(n-1+j)(n+j)} > 0.$$

Thus a_j is log-convex. The sequence $1 + a_j$ is also log-convex. Indeed, if positive adjacent values u, v, w satisfy $v^2 \leq uw$, then

$$(1+u)(1+w) - (1+v)^2 = (uw - v^2) + (u + w - 2v) \geq 0.$$

It follows that

$$h_j := \frac{r_j}{1 + a_j}$$

is log-concave on $0 \leq j \leq n + 1$. Consequently the set of indices for which $h_j > 1$ is an interval of integers: equivalently, the positive superlevel set of the discrete concave sequence $\log h_j$ is an interval. The negative coefficients of P_n are precisely those indexed by this set. They therefore form a single block. The coefficients below degree $n - 2$ are positive, and the top coefficient (where $r_{n+2} = 0$) is $1 + 2^{n+3} > 0$. Deleting any zero coefficients cannot add a sign change. Thus P_n has at most two coefficient sign variations, and Descartes' rule gives at most two positive roots counted with multiplicity.

5 Odd-case existence and completion of the proof

For odd n , the middle-interval calculation gives $Q_n(-1/2) > 0$, and $Q_n(0) = 1$. When $n = 3$, direct substitution in (1) gives

$$D_3(-1/5) = -\frac{589}{390625} < 0.$$

For every odd $n \geq 5$, another exact substitution gives

$$9^{n+1}D_n(-1/3) = 4^n - 3 \cdot 5^n + 18 \cdot 3^n < 0. \quad (10)$$

Indeed, after division by 5^n , the positive part is

$$\left(\frac{4}{5}\right)^n + 18 \left(\frac{3}{5}\right)^n,$$

which decreases with n and equals $5398/3125 < 3$ at $n = 5$. Because D_n and Q_n have the same sign away from zero, each odd n has a negative interior witness in $(-1/2, 0)$ between two positive endpoint values. The intermediate value theorem gives two distinct roots. They saturate the Descartes multiplicity bound from the preceding section, so they are the only roots in that interval and both are simple. This also shows that the odd-case coefficient sequence has exactly two sign variations.

The affine substitution used for T_n preserves algebraic multiplicity. The rational substitution used for P_n has nonzero derivative for $t > 0$, and its clearing factor is nonzero there, so it also preserves algebraic multiplicity. Finally, $D_n = x^2 Q_n$ with $x^2 \neq 0$ away from zero. The left interval, the closed middle interval, $(-1/2, 0)$, zero, and the positive axis exhaust the real line. Theorem 1 follows.

Remark 1. *The lower boundary on n is not cosmetic. Formula (1) gives*

$$D_1(x) = x^2(x^2 + 4x + 6) = x^2((x + 2)^2 + 2),$$

so B_1 has no nonzero real domination root. No assertion about $n = 0$ or about the nonreal zeros is made here.

Computational and AI assistance

An OpenAI language model assisted with symbolic derivations, exact-arithmetic checks, proof organization, and manuscript drafting. The all-parameter proof above is analytic and does not infer

an infinite statement from finite data. As corroboration, exact computations enumerated dominating sets for $1 \leq n \leq 6$, checked the transformed coefficients for $2 \leq n \leq 40$ and for $n = 67, 68$, and used rational Sturm chains for ten selected values up to $n = 34$. A separate fresh model-context audit independently reconstructed the normalized coefficient formulas, checked the support endpoints and the odd log-concavity argument, and verified that the substitutions preserve the multiplicity bounds. Such an audit is neither external peer review nor proof-assistant certification.

References

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